MUREN AI: Human-Object Interaction (HOI) Detection Suite

Complete Reference: 117 Action Verbs, 80 Objects & 600 HOI Triplet Capabilities

Executive Summary: Model Architecture & Detection Capabilities

This reference manual catalogs all 80 object classes, 117 interaction action verbs, and 600 validated Human-Object Interaction (HOI) triplets detected by the MUREN relational transformer system.

117
Action Verbs
Daily, sports, postures

80
Object Classes
COCO 80 categories

600
HOI Triplets
<Human, Verb, Object>

64
Query Slots
MUREN Transformer Qs

1. Core Technical Specifications

- **Backbone Architecture:** ResNet-50 (Pretrained on ImageNet-1K), multi-scale feature projections
- **Relation Engine:** Context-Aware Decoupled Transformer Encoder (6 layers) & Decoder (6 layers)
- **Query Mechanism:** 64 learned interactive relation queries (simultaneous human-object reasoning)
- **Inference Checkpoint:** hico-best.pth (HICO-DET Benchmark, 301.5 MB weights)
- **Detection Modalities:** Single Image (Still), Video Stream Tracking, Live Webcam, and Batch Audits

2. Overview of Action Verbs (117 Total) by Functional Group

Posture & Body Motion (13)
sit_on, sit_at, stand_on, stand_under, lie_on, walk, run, jump, slide, straddle, hop_on, exit, board

Daily Manipulation & Handling (16)
hold, carry, lift, pull, push, drag, open, pack, pick, pick_up, move, adjust, assemble, break, repair, lose

Communication, Tech & Media (12)
talk_on, type_on, text_on, watch, inspect, read, sign, point, check, control, operate, turn

Food Preparation & Dining (16)
eat, eat_at, drink_with, sip, cook, cut, cut_with, peel, pour, stir, serve, feed, lick, smell, toast, make

Vehicles, Transit & Mobility (11)
ride, drive, park, fly, launch, race, row, sail, direct, load, hose

Sports, Athletics & Recreation (11)
hit, kick, throw, catch, dribble, swing, spin, lasso, wield, tag, release

Personal Care, Grooming & Animals (16)
wear, tie, zip, brush_with, wash, dry, clean, groom, pet, herd, milk, shear, hunt, hug, kiss, greet

Household, Crafts & Utility (22)
paint, flush, fill, flip, grind, light, pay, buy, scratch, set, squeeze, stab, stick, teach, train, wave, blow, block, stop_at, no_interaction

3. Complete Master Alphabetical Index of Action Verbs (Part 1: Verbs 1 to 58)

#	Verb Label	Typical Context Objects	#	Verb Label	Typical Context Objects
1	adjust		30	flush	
2	assemble		31	fly	
3	block		32	greet	
4	blow		33	grind	
5	board		34	groom	
6	break		35	herd	
7	brush_with		36	hit	
8	buy		37	hold	
9	carry		38	hop_on	
10	catch		39	hose	
11	chase		40	hug	
12	check		41	hunt	
13	clean		42	inspect	
14	control		43	install	
15	cook		44	jump	
16	cut		45	kick	
17	cut_with		46	kiss	
18	direct		47	lasso	
19	drag		48	launch	
20	dribble		49	lick	
21	drink_with		50	lie_on	
22	drive		51	lift	
23	dry		52	light	
24	eat		53	load	
25	eat_at		54	lose	
26	exit		55	make	
27	feed		56	milk	
28	fill		57	move	
29	flip		58	no_interaction	

4. Complete Master Alphabetical Index of Action Verbs (Part 2: Verbs 59 to 117)

#	Verb Label	#	Verb Label
59	**open**	89	**slide**
60	**operate**	90	**smell**
61	**pack**	91	**spin**
62	**paint**	92	**squeeze**
63	**park**	93	**stab**
64	**pay**	94	**stand_on**
65	**peel**	95	**stand_under**
66	**pet**	96	**stick**
67	**pick**	97	**stir**
68	**pick_up**	98	**stop_at**
69	**point**	99	**straddle**
70	**pour**	100	**swing**
71	**pull**	101	**tag**
72	**push**	102	**talk_on**
73	**race**	103	**teach**
74	**read**	104	**text_on**
75	**release**	105	**throw**
76	**repair**	106	**tie**
77	**ride**	107	**toast**
78	**row**	108	**train**
79	**run**	109	**turn**
80	**sail**	110	**type_on**
81	**scratch**	111	**walk**
82	**serve**	112	**wash**
83	**set**	113	**watch**
84	**shear**	114	**wave**
85	**sign**	115	**wear**
86	**sip**	116	**wield**
87	**sit_at**	117	**zip**
88	**sit_on**		

5. Complete Catalog of 80 Detectable Object Classes (COCO / HICO-DET)

ID	Object Name	ID	Object Name	ID	Object Name	ID	Object Name
1	person	21	elephant	41	wine glass	61	dining table
2	bicycle	22	bear	42	cup	62	toilet
3	car	23	zebra	43	fork	63	tv
4	motorcycle	24	giraffe	44	knife	64	laptop
5	airplane	25	backpack	45	spoon	65	mouse
6	bus	26	umbrella	46	bowl	66	remote
7	train	27	handbag	47	banana	67	keyboard
8	truck	28	tie	48	apple	68	cell phone
9	boat	29	suitcase	49	sandwich	69	microwave
10	traffic light	30	frisbee	50	orange	70	oven
11	fire hydrant	31	skis	51	broccoli	71	toaster
12	stop sign	32	snowboard	52	carrot	72	sink
13	parking meter	33	sports ball	53	hot dog	73	refrigerator
14	bench	34	kite	54	pizza	74	book
15	bird	35	baseball bat	55	donut	75	clock
16	cat	36	baseball glove	56	cake	76	vase
17	dog	37	skateboard	57	chair	77	scissors
18	horse	38	surfboard	58	couch	78	teddy bear
19	sheep	39	tennis racket	59	potted plant	79	hair drier
20	cow	40	bottle	60	bed	80	toothbrush

6. Object Groups & Category Distribution

- **People & Animals (11):** person, bird, cat, dog, horse, sheep, cow, elephant, bear, zebra, giraffe
- **Vehicles & Transport (8):** bicycle, car, motorcycle, airplane, bus, train, truck, boat
- **Outdoor Infrastructure (5):** traffic light, fire hydrant, stop sign, parking meter, bench
- **Personal Accessories (5):** backpack, umbrella, handbag, tie, suitcase
- **Sports Equipment (10):** frisbee, skis, snowboard, sports ball, kite, baseball bat, baseball glove, skateboard, surfboard, tennis racket
- **Kitchen Utensils (7):** bottle, wine glass, cup, fork, knife, spoon, bowl
- **Food & Edibles (10):** banana, apple, sandwich, orange, broccoli, carrot, hot dog, pizza, donut, cake
- **Indoor Furniture (6):** chair, couch, potted plant, bed, dining table, toilet
- **Electronics & Computing (6):** tv, laptop, mouse, remote, keyboard, cell phone
- **Appliances & Household (12):** microwave, oven, toaster, sink, refrigerator, book, clock, vase, scissors, teddy bear, hair drier, toothbrush

7. Common Real-World HOI Triplets Detected by MUREN

Domain	Primary Action Verb	Target Object	Canonical Interaction Triplet <Subject, Verb, Object>
Office / Work	**type_on**	laptop	**<person, type_on, laptop>**
Office / Work	**talk_on**	cell phone	**<person, talk_on, cell phone>**
Office / Work	**read**	book	**<person, read, book>**
Office / Work	**sit_at**	dining table	**<person, sit_at, dining table>**
Urban Mobility	**ride**	bicycle	**<person, ride, bicycle>**
Urban Mobility	**drive**	car	**<person, drive, car>**
Urban Mobility	**board**	bus	**<person, board, bus>**
Urban Mobility	**hop_on**	motorcycle	**<person, hop_on, motorcycle>**
Sports & Fitness	**hit**	sports ball	**<person, hit, sports ball>**
Sports & Fitness	**kick**	sports ball	**<person, kick, sports ball>**
Sports & Fitness	**surf**	surfboard	**<person, ride, surfboard>**
Sports & Fitness	**swing**	tennis racket	**<person, swing, tennis racket>**
Dining & Kitchen	**eat**	pizza	**<person, eat, pizza>**
Dining & Kitchen	**drink_with**	cup	**<person, drink_with, cup>**
Dining & Kitchen	**cut_with**	knife	**<person, cut_with, knife>**
Dining & Kitchen	**stir**	bowl	**<person, stir, bowl>**
Living & Home	**lie_on**	couch	**<person, lie_on, couch>**
Living & Home	**sit_on**	chair	**<person, sit_on, chair>**
Living & Home	**watch**	tv	**<person, watch, tv>**
Living & Home	**open**	refrigerator	**<person, open, refrigerator>**
Animal Care	**pet**	dog	**<person, pet, dog>**
Animal Care	**feed**	cat	**<person, feed, cat>**
Animal Care	**ride**	horse	**<person, ride, horse>**
Personal Items	**carry**	backpack	**<person, carry, backpack>**
Personal Items	**wear**	tie	**<person, wear, tie>**

8. Operational Tuning & Confidence Threshold Recommendations

- **Default Action Threshold:** 0.20 (20%) - Balances high recall across subtle actions (e.g. looking at, inspecting)
- **Default Object Threshold:** 0.50 (50%) - Ensures high precision bounding boxes from DETR detector queries
- **Max Interaction Queries:** Default: 5 per image (Prevents visual clutter; adjustable up to 20 in UI)
- **Video Sampling Frequency:** Sample 1 frame every 3-5 frames (preserves temporal continuity with fast processing)
- **Real-Time Webcam Interval:** Recommended 150ms-250ms stream packet interval for low-latency CPU inference

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